

Tutorial No.5

Q.1) If,

$$V_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad V_2 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \quad \text{Use standard inner product of } \mathbb{R}_2$$

And find a) $\langle V_1, V_2 \rangle$ b) $\langle V_1, V_3 \rangle$ c) $\langle V_2, V_3 \rangle$ d) $\|V_1\|$ e) $\|V_2\|$ f) $\|V_3\|$

Q.2) Let P (t) be vector space of polynomials and for f(t), g(t) in P(t)

$$\text{inner product} \quad \langle f(t), g(t) \rangle = \int_{-1}^1 f(t)g(t)dt$$

$$\text{If } f(t) = t + 2 \quad \& \quad g(t) = 3t - 2 \quad \& \quad h(t) = t^2 - 2t - 3$$

then find $\langle f(t), h(t) \rangle$ & $\|g\|$

Q.3) If,

$$V_1 = \begin{bmatrix} 1 & 1 \\ -2 & 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} -2 & 2 \\ -2 & 2 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$$

Use standard inner product of $M_{2 \times 2}(R)$

Find a) $\langle V_1, V_2 \rangle$ b) $\langle V_1, V_3 \rangle$ c) $\langle V_2, V_3 \rangle$ d) $\|V_1\|$ e) $\|V_2\|$ f) $\|V_3\|$

Q.4] Find the value of k for which the vectors u = (k, -1, k) and v= (0 , 1, k) are orthogonal.

Q.5] Find Projection of vector (0, 2, 4) along the vector (3, 3, 3)

Q.6] Find angle between two vectors $V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $V_2 = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$